

Operating safely, ethically and legally

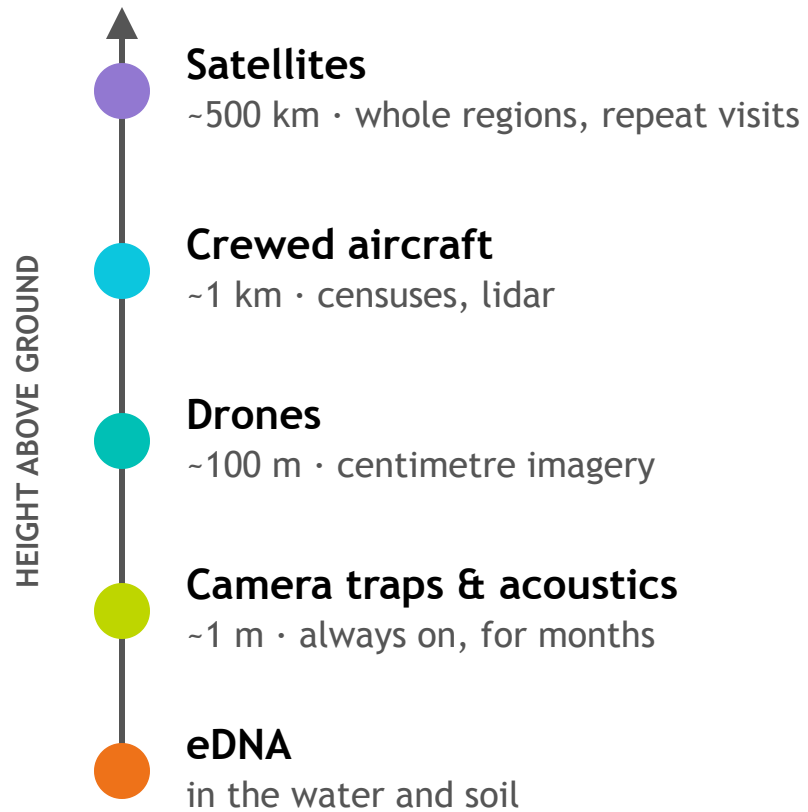
Experiential AI & Ecology

Module 3: multimodal ecosystem sensing

Steve Bullock · University of Bristol



Every capability is a responsibility



...and the AI that turns it all
into counts and maps.

Three lenses:

Safe

nobody, human or animal, gets hurt

Ethical

what you *should* do, not just what you *can*

Legal

the rules where and how you work

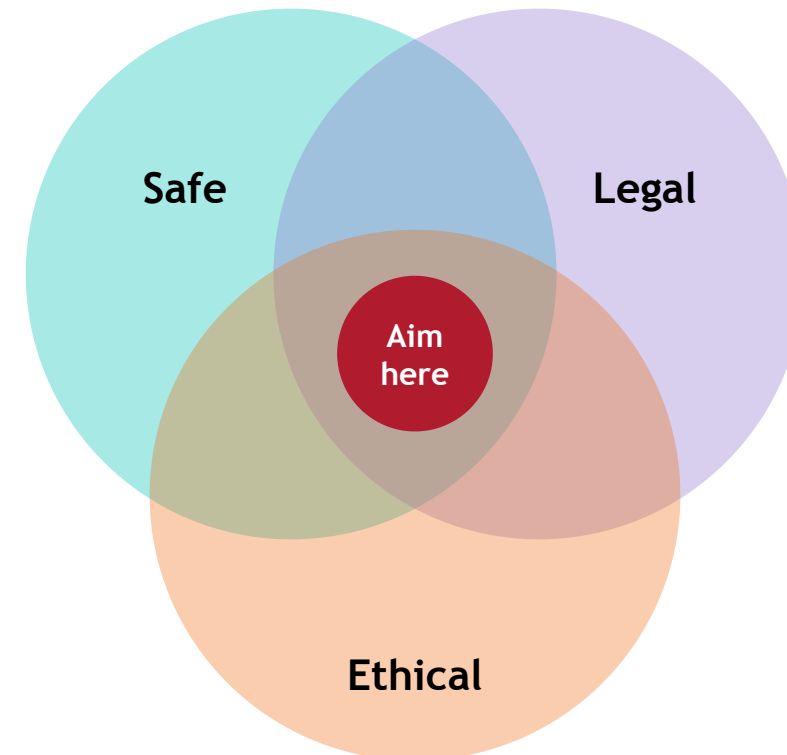
Safe is not the same as compliant

Safety = reducing the actual risk of harm

Compliance = following the applicable rules, standards and procedures

You can be **compliant but unsafe**, if the rules are incomplete or applied mechanically.

You can be **safe but not compliant**, if you break a formal requirement even when the real risk is low.

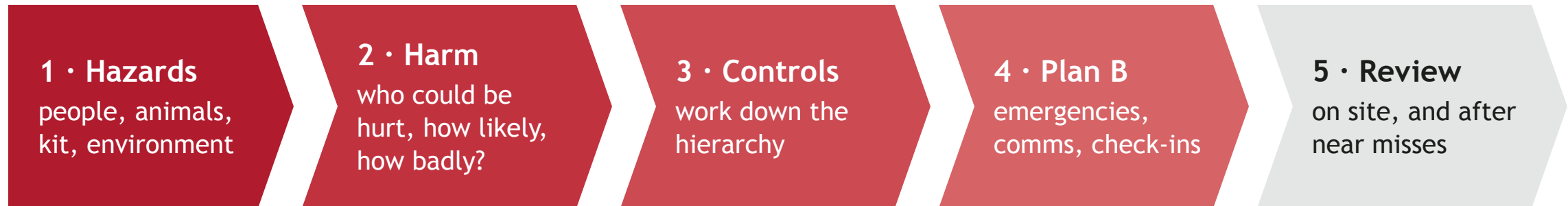


Safe

Nobody, human or animal, gets hurt



Assess the risk before you go



Use your institution's fieldwork risk assessment and the course's field safety plan. They're there to make you think, not to collect signatures.

Air risk

what else is in the sky?

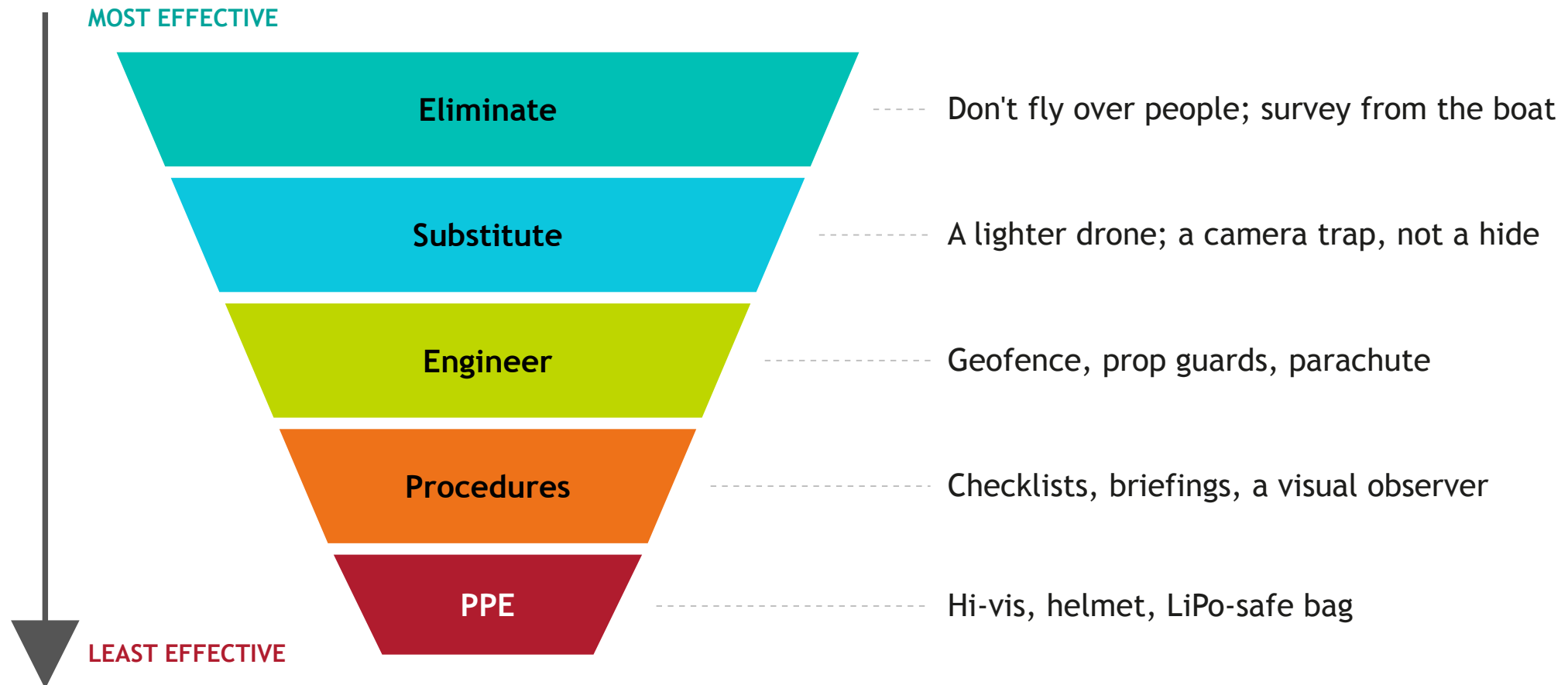


Ground risk

who and what is underneath?



The hierarchy of controls



Sources and further reading: [Principles of prevention, MHSWR 1999 Sch. 1](#) · [HSE: risk assessment](#)

Procedures, every time

Two golden rules

1

Develop and use procedures and checklists.
Otherwise, at some point, you will forget something critical.

2

Never have the propellers fitted and the battery connected at the same time, unless you are about to take off.

Field lessons

- ✓ Do your homework, with local people and institutions
- ✓ Do the paperwork before you go
- ✓ Recce the site if you can
- ✓ Clear test plan; beware scope creep
- ✓ Plan rest days
- ✓ Don't count on one deployment
- ✓ Keep structured records

Who else is at risk?

The animals

Drones can disturb wildlife in ways a person on foot would not: species, altitude, approach and noise all matter.

People nearby

Communities, landowners, other airspace users, your own team.

People downstream

If a detector misses a poacher or overcounts a population, someone acts on a wrong answer.

Who checks the model?

Ethical

What you should do, not only what you can



Technology is not ethically neutral

BUILT TO...

A thermal drone finds animals at night

Camera traps record wildlife

Acoustic recorders capture birdsong

Precise locations support conservation

50 cm imagery resolves one animal

An open model tracks animals

...BUT ALSO

→ finds people

→ photograph hunters, herders and villagers

→ capture speech

→ guide poachers and collectors

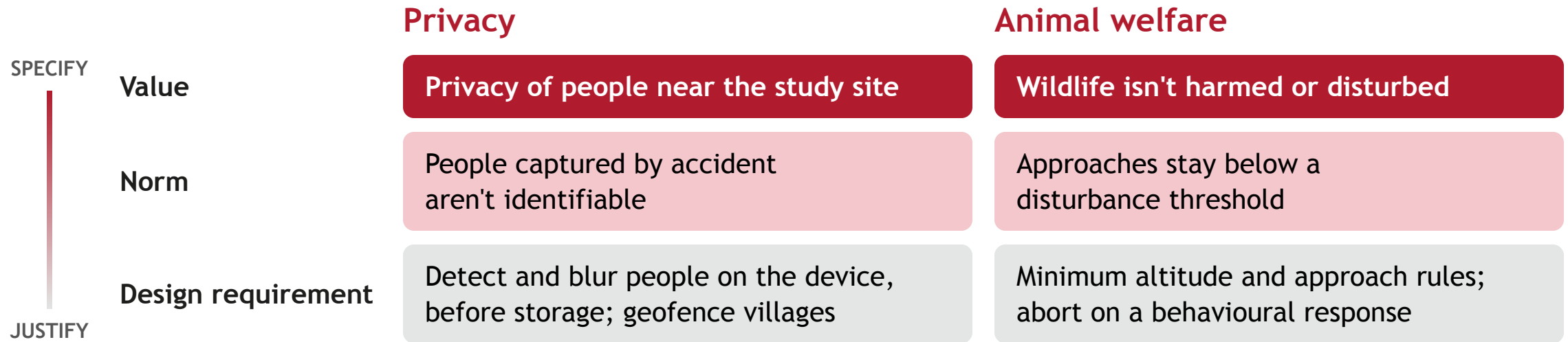
→ resolves homes and vehicles

→ tracks people

Start with **Sandbrook et al. (2021)**: principles for socially responsible conservation monitoring.

Design values in from the start

Value-sensitive design (Friedman & Hendry 2019): who are the stakeholders, which values are at stake, and how do you build them in?



Sources and further reading: [Friedman & Hendry 2019, Value Sensitive Design](#) · [van de Poel 2013, Translating values into design requirements](#)

Capability caution



Ask of each capability: **do we need it, and how could it be misused?**

1. **Context of use:** where, by whom, under whose rules?
2. **Privacy:** what does it record that it doesn't need?
3. **Jobs and human skills:** whose work does it change or replace?
4. **Safety, security and misuse:** who could use it, or its data, for harm?
5. **The future:** what will it make possible in ten years, in other hands?

Sources and further reading: [Cawthorne & Devos 2020](#), [Capability caution in UAV design](#) · [Stilgoe et al. 2013](#), [Responsible innovation](#)

Share data fairly

FAIR is about data

F

Findable

A

Accessible

I

Interoperable

R

Reusable

CARE is about people

CCollective
benefit**A**Authority to
control**R**

Responsibility

E

Ethics

- Who benefits from your data and models?
- Do the communities where you collect gain anything?
- Do sensitive species' locations need generalising before release?
- Does your work need **ethics review**?
Animals, people (even by accident), fieldwork abroad, eDNA, Indigenous lands or data

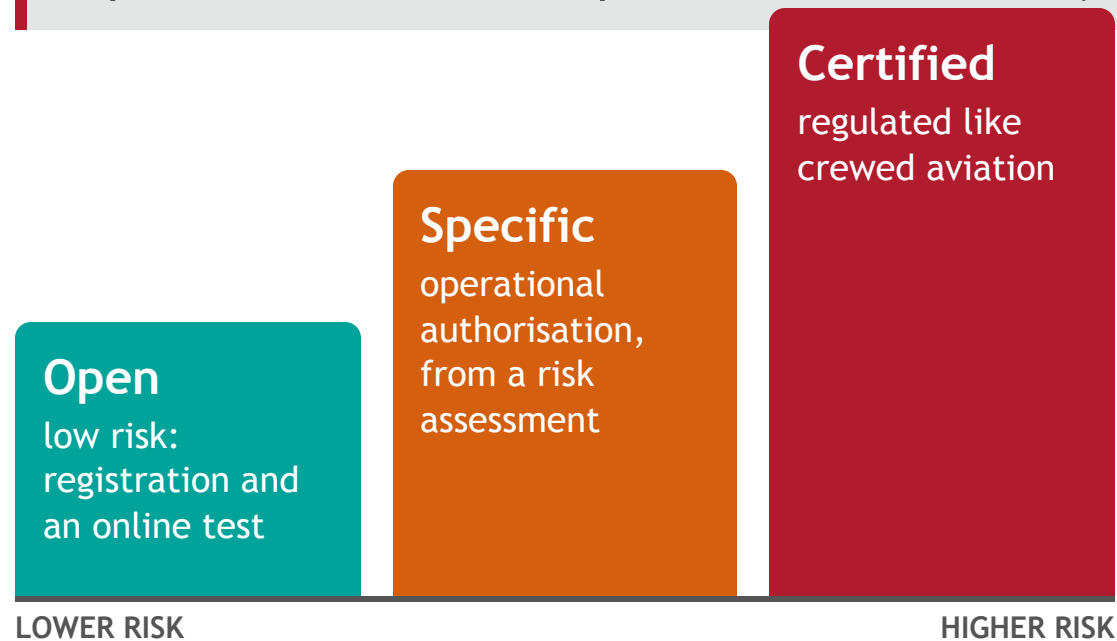
Legal

Mostly drones: flying puts you under aviation law everywhere



UK and EU: risk-based, not purpose-based

Correct as of October 2026, and my interpretation. Not legal advice. It's up to you to operate within the requirements of where you fly and of your own institution.

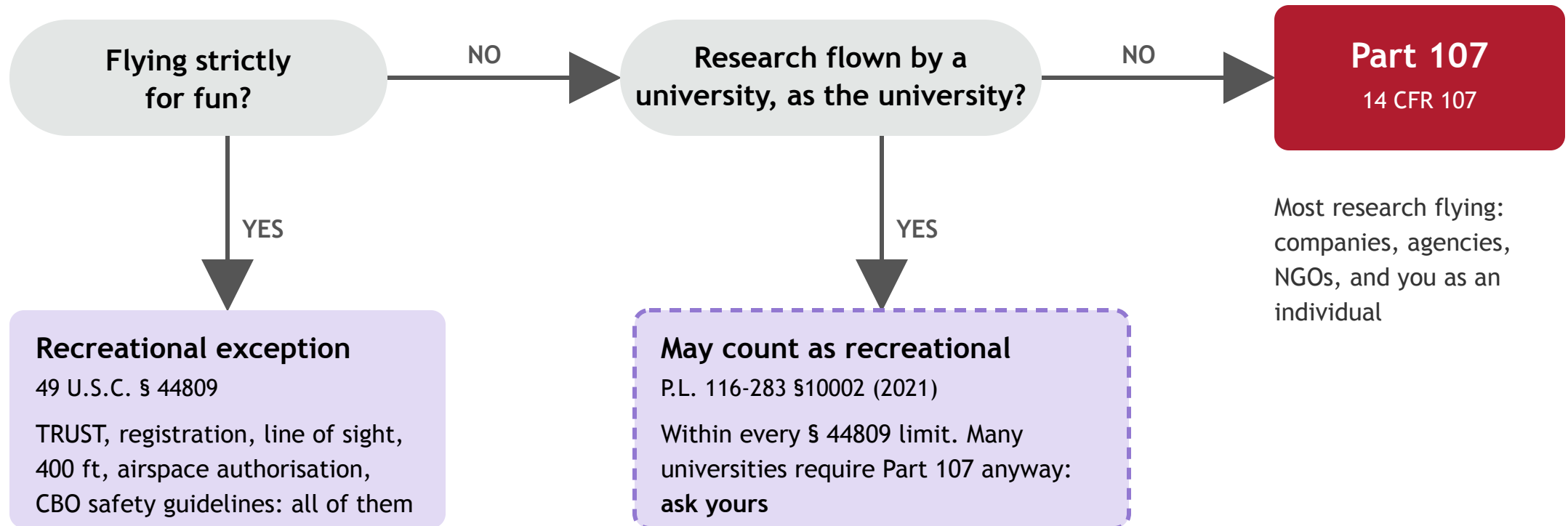


No recreational/commercial distinction.
Same drone, same place, same rules.

UK (CAA): free **Flyer ID** from 100 g since January 2026, plus an **Operator ID**

EU (EASA): register and pass the free **A1/A3** online exam with your national authority

US: Part 107 or the recreational exception



Check before you fly, and beyond aviation

Everywhere: some countries restrict drones heavily, need import permits, or ban them in protected areas. Check the aviation authority, the protected-area authority and local partners well before you travel, and an airspace map on the day.

Privacy

People in images
and audio are
personal data

Wildlife

Disturbance
offences, research
permits

Genetic resources

eDNA and the
Nagoya Protocol

Export controls

Thermal cameras,
some drones and
software

Data licences

Commercial
imagery terms

Before the live session

1

Drone basics

~30 min, if you might fly

UK Flyer ID · EU A1/A3 · US TRUST

Explore an airspace map for home
and your field site

2

Read

~10 min

Safe, ethical, legal: with links to
go further

3

Capability audit canvas

~20 min

Fill in parts 1-5 for your own
system and upload it

**Part 6, the red team,
happens live**

Be specific: *"thermal camera at 120 m picks out people in the village by the reserve"*
beats *"privacy"*.



See you in the live session

steve.engineer/teaching/imageomics2026

